AEROSPACE DEFENSE SIMULATION REPORT

PROJECT-12

Boeing AH-64E Apache Guardian Border Defense War Simulator

Lead Architect:	Mr ELONIKHIL (mr elonikhil)
Airframe Model:	Boeing AH-64E Apache Guardian
Subsystem Count:	4 Integrated Avionics Suites
CAD Parts Count:	1,480 Precision Mated Solid Bodies
Adversary Units:	12 Unique Threat Attack Profiles
Physics Paradigm:	6-DOF Rigidbody + Anti-Gravity
3D Engine Pipeline:	PlayCanvas JS / WebGL & Blender 5.2
License:	MIT Open Source
Release Date:	October 2026

1. Executive Summary

Project 12 represents an ultra-high-fidelity, zero-dependency browser-based combat vehicle simulation and 3D CAD modeling suite. Built on a 6-DOF Rigidbody physics paradigm, the software faithfully replicates the Boeing AH-64E Apache Guardian attack helicopter operating over the mountainous terrain of the Kashmir Line of Control (LOC) around Forward Operating Base (FOB) Uri.

The vehicle replaces mechanical rotor physics with an advanced anti-gravity propulsion core capable of delivering continuous vertical thrust counteracting gravity, dynamic aerodynamic drag damping, and automatic center-of-mass balancing against asymmetrical weapon stores.

> **Zero-Dependency Engineering Mandate:** All 3D vector calculations, matrix transformations, PID anti-windup clamping, cubic Hermite spline interpolation, 90-degree conical radar sweep dot products, and procedural Web Audio threat tone synthesis are implemented natively in pure code with zero third-party plugins.

2. Subsystem Architecture Specifications

Subsystem	Primary Module	Core Mathematical Functions & Avionics Roles
System 1: GNC Engine	`GNCSystem.js`	- Discrete 3D PID hover hold with anti-windup clamping. - Low-pass filtered derivative term (alpha = 0.25). - 3-ray predictive TFR array (0°, 30°, 45° lookahead, 50m-300m range). - Cubic Hermite spline target altitude smoother (+15.0m clearance).
System 2: EW & Radar Suite	`ElectronicWarfareSuite.js`	- 360° RWR azimuth tracking normalized to [-180°, +180°]. - Tri-state threat alerts: SEARCH (>500m), TRACK (<=500m), LOCK (missile). - Web Audio API procedural synthesis: 1Hz (SEARCH), 5Hz (TRACK), continuous (LOCK). - Automated CMDS: 0.25s time-locked dispenser of active thermal Flares & reflective Chaff.

Subsystem	Primary Module	Core Mathematical Functions & Avionics Roles
		- Intercept algorithm overriding incoming missile target transforms.
System 3: Tactical Data Link	`TacticalDataLink.js`	- Link 16 military data bus with distance-dependent latency (0.1s - 0.5s). - Simulated radio-electronic position distortion. - FCR 90° conical volumetric sweep (+/- 45° azimuth, 1000m range). - 3-factor Threat Index Weight: 45% Proximity, 35% Threat Class, 20% Closure Velocity. - Priority sort of top 16 contacts with Index 0 Primary Weapon Lock.
System 4: Stores Management	`StoresManagementSystem.js`	- 4 wing hardpoints (Hellfire, Hydra pods) + 30mm central ammo drum. - Dynamic Gross Mass recalculation upon weapon discharge. - Weighted-average Center of Mass (CoM) physical vector offset. - Asymmetrical rolling torque simulation (tau = R_CoM x -Mg). - 5-page military MFD text stream (MENU, GNC_TFR, EW_RWR, TAC_LINK, SMS_STORES).
Master Controller	`AntiGravityHelicopterMaster.js`	- Rigidbody flight physics integrator. - Anti-gravity vertical thrust, aerodynamic drag, angular rate damping. - Automatic roll trim counteracting asymmetrical ordnance loadouts. - Pilot flight input translator (cyclic, collective, anti-torque rudder).

3. Operational Combat Capabilities

3.1 Weapons Suite

- **M230 30mm Chain Gun:** Continuous cyclic fire, dual-layer illuminated tracers (incandescent white core + golden corona), dynamic muzzle flash, and physical spent brass casing ejection.

- **AGM-114L Hellfire:** Top-attack loft trajectory climbing +52m before diving into target top armor.
- **Hydra 70 Rockets:** High-explosive folding-fin aerial rocket salvos.
- **1-Shot Super Smash ([X]):** Salvo obliterating all 12 hostile units simultaneously in synchronized cluster strikes.

3.2 12 Hostile Border Adversary Units

The theater models 12 distinct enemy threat types: HQ-17A Mobile SAM Battery, Al-Khalid Main Battle Tank, ZSU-23-4 Shilka Quad AAA, Insurgent Technical Pickup (12.7mm DShK), RPG-7 Mountain Ambush Squad, 120mm Heavy Mortar Dugout, Anza Mk-II / QW-2 MANPADS, Shahed-136 Kamikaze Drone, Mi-35 Hind Gunship, Riverine Patrol Boat, Early Warning Phased-Array Radar Bunker, and Mountain Ridge Sniper Redoubts.
